

ML

1. Supervised learning, regression
 1. $f(X) = P(Y|X)$
2. dataset X , marginal distribution, $\mu(x) = P(X)$
3. defined so called regression function, $\eta(x) = E[Y|X = x]$, read: expectation of the prediction Y given $X = x$.
4. In the special case of classification, the regression function can be rewritten as
$$\eta(x) = 0 \cdot P(Y = 0, X = x) + 1 \cdot P(Y = 1|X = x) = P(Y = 1|X = x),$$
but this is just regression! or being reduced to a regression problem.
5. If the expectations of the predictions, $\eta(x)$ is close to 0.5, i.e. if η is close to 1, then classifying x is easy if η is close to 0.5, then classifying x is difficult.
6. The probability distribution $P(X, Y)$ is uniquely determined by μ and η .
7. Intuition (discrete case): We can rewrite $P(X = x, Y = 1) = P(Y = 1|X = x)P(X = x) = \eta(x)\mu(x)$ and similarly,
$$P(X = x, Y = 0) = P(Y = 0|X = x)P(X = x) = (1 - \eta(x))\mu(x)$$
8. So we can express the probability of any event (x, y) in terms of η and μ .
9. For formal proof for the general case: see the book of Devorye, Gyofri, Lugosi, first pages.
10. Explicit form of Bayes classifier under 0-1 loss.
 1. Note: now assume the classifier $f(X) = P(Y|X = x)$ gives either 0 or 1 or $\{0, 1\}$.
 2. The risk of a classifier under the 0-1 loss counts how often the classifier fails, that is
$$R(f) = E(l(X, Y, f(X))) = E(\mathbf{1}_{f(X) \neq Y}) = P(f(X) \neq Y).$$
 3. The bayes classifier f^* was defined as the classifier that minimizes the **true** risk. This is a hypothetical classifier function.
11. Note: Linear regression
 1. Why not just use matrix inversion? because matrix inversion scales badly with number of features d . With n rows and d features compute is $O(d^3 + nd^2)$ and overhead space is $O(d^2)$. Iterative (gradient descent) methods are better for larger datasets. For small d matrix inversion scales just as well as gradient descent.
12. Empirical Risk Minimization (ERM)
 1. Define a set $\mathcal{F}$ of functions from $X \rightarrow Y$.
 2. Within these functions, choose one that has the smallest empirical risk.
 1. $f_n = \arg \min_{f \in \mathcal{F}} R_n(f)$
 2. (if minimum does not exist, only infimum exists, then it will be that converges to a function f_n . The function sequence classifier functions may not even be unique, but in practice it does not pose any problems).
 3. Estimation vs approximation (model) error
 4. Underfitting means $\mathcal{F}$ is too small:
 1. small estimation error.
 2. large model error (approximation error).
 5. Overfitting means $\mathcal{F}$ is too large:
 1. large estimation error.
 2. small model error (approximation error).
 6. Bias-variance trade-off
 1. Bias term: how much difference between f_n and f^* (same flavour as model error).
 2. Variance term: the variance of the RV f_n (estimation error).
 7. ERM is a straightforward learning principle.
 8. The key to success/failure of ERM is to choose a 'good' function class $\mathcal{F}$.
 9. Finding the minimizer of the 0-1 loss can be challenging as it is often NP-hard. In practice, we use convex relaxation of the 0-1 loss function, see below.
 10. Approach:
 1. Define regularized risk: $R_{reg,n}(f) = R_n(f) + \lambda \Omega(f)$. Here $\lambda > 0$ is called a regularizer constant.
 2. Then choose $f \in \mathcal{F}$ to minimize the regularized risk.
 3. Regularizer, $\Omega : \mathcal{F} \rightarrow \mathbb{R}_{>0}$, that measures how 'complex' a function is,
 4. Examples:
 1. $\mathcal{F}$ = polynomials, Ωf = degree of the polynomial f .

2. $\mathcal{F}$ = differential functions, $\Omega(F)$ = maximal slope.

5. if λ very small: overfitting, high bias error.

6. if λ very large: underfitting, high variance error.

13. when we talk about a distribution

1. model
2. parameter
3. random variable

14. distribution models (based on study order):

1. Bernoulli, RV: $X = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{with probability } q=1-p \end{cases}$
2. Binomial, RV: getting exactly k successes in n independent trials.
3. Normal/Gaussian, RV: continuous real number.
4. Poisson, RV: number of events in a fixed interval of time (or space).
5. Exponentials, RV: the amt of time until a next event occurs (success).
6. Gamma, RV: the wait time until the k -th event occurs.
7. Chi-squared, RV: sum of squared of independent standard normal RVs.
8. Beta, RV: continuous value between 0 and 1, i.e. $[0, 1] \in \mathbb{R}$.

15. Bayesian decision rule

1. Given the height of a person we want to predict the gender.
2. RV: Y: male or female, RV: X: height.
3. Code:

```
# -*- coding: utf-8 -*-
"""
Created on Sat Apr 11 20:17:07 2026

@author: reina
"""

import numpy as np
import matplotlib.pyplot as plt

# probability density function or likelihood
def gaussian_pdf(x, mu, var):
    a = 1 / np.sqrt(2 * np.pi * var)
    b = - (x - mu)**2 / var
    return a * np.exp(b)

# RV: X: height
# RV: Y: male or female
# want to predict P(Y | X)

# Define likelihood
x = np.linspace(0, 220, num=100)
mu1 = 175; var1 = 50
mu2 = 162; var2 = 50
likelihood1 = gaussian_pdf(x, mu1, var1)
likelihood2 = gaussian_pdf(x, mu2, var2)

# Define priors
prior1 = 0.3
prior2 = 1 - prior1

# Compute marginal
marginal = likelihood1 * prior1 + likelihood2 * prior2

# Compute posteriors
posterior1 = likelihood1 * prior1 / marginal
posterior2 = likelihood2 * prior2 / marginal

fig = plt.figure(figsize=(16, 12))

# Plot likelihood
ax1 = fig.add_subplot(221)
ax1.plot(x, likelihood1, color='b')
ax1.plot(x, likelihood2, color='r')
ax1.set_xlim(130, 200)
ax1.set_ylabel('Likelihood, P(X | Y)')

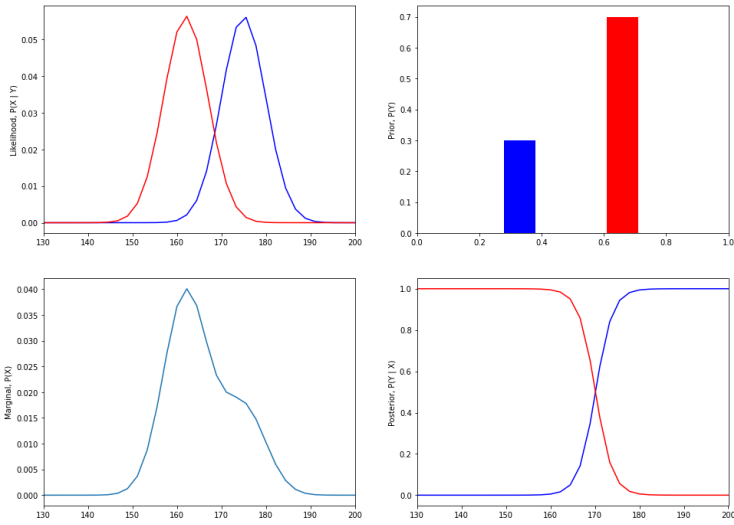
# Plot priors
ax2 = fig.add_subplot(222)
ax2.bar(0.33, prior1, width=0.1, color='b')
ax2.bar(0.66, prior2, width=0.1, color='r')
ax2.set_xlim(0, 1)
ax2.set_ylabel('Prior, P(Y)')
```

```

# Plot marginal
ax3 = fig.add_subplot(223)
ax3.plot(x, marginal)
ax3.set_xlim(130, 200)
ax3.set_ylabel('Marginal, P(X)')

# Plot posterior
ax4 = fig.add_subplot(224)
ax4.plot(x, posterior1, color='b')
ax4.plot(x, posterior2, color='r')
ax4.set_xlim(130, 200)
ax4.set_ylabel('Posterior, P(Y | X)')

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References:

1. Statistical Machine Learning, Tübingen, 2024.